# Lakshay Chauhan

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## Education

**Indraprastha Institute of Information Technology (IIIT) Delhi** Jul 2021 – Jun 2025
*Bachelor of Technology in Computer Science and Engineering* *Delhi, India*

## Work Experience

**Member of Technical Staff** Jul 2025 – Present
*ZL Technologies, Inc | **Website*** *Hyderabad, India*

- Developed task synchronization and bulk parallel processing mechanisms across clusters within ZL's analytics module, achieving over 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection for large-scale data loading workflows processing millions of items.

**Research Assistant | *Gitlab Project Organization*** May 2024 – May 2025
*Networks and Systems Security Lab - IIITD | **Website*** *Delhi, India*

- Researched mechanisms to mitigate video streaming quality degradation during fallback from QUIC to TCP caused by UDP blocking.
- Implemented QUIC-based notification signaling and kernel congestion window state reuse to bypass TCP slow start, reducing buffering and improving user experience during fallback transitions.

## Projects

**Minimal Compiler Infrastructure | *Repository*** May 2024 – Present

- ***Key Skills:*** *Compiler Infrastructure, Design and Optimization, Rust, FASM, WASM, Graphviz*
- Developed a minimal compiler infrastructure inspired by LLVM. It converts source code into an optimized control flow graph (CFG) and translates it into target-specific assembly code.
- Implemented CFG optimization passes, such as identifier validation, constant folding, and CFG simplification, ensuring high-performance code generation.

**Exploring Compiler Optimizations | *Repository*** Nov 2024 – Feb 2025

- ***Key Skills:*** *Compiler Optimization, Compiler Passes, Garbage Collectors, Type Safety, LLVM, C, C++*
- Developed LLVM-based compiler and runtime systems including a null pointer dereference detection pass, a conservative bump-pointer garbage collector for C with root scanning, and memory safety mechanisms enforcing spatial and weak type safety through bounds tracking and invalid pointer prevention.

**Garbage Collector for C | *Repository*** Feb 2025 – Present

- ***Key Skills:*** *Runtime Systems, C, Garbage Collection, Memory Management, Systems Programming*
- Developed a conservative mark-and-sweep garbage collector for C with an 8-byte aligned bump allocator and segmented heap design.
- Implemented stack, heap, and global-root scanning for automatic memory reclamation.
- Designed page-level memory management using `mmap`, `mprotect`, and `madvise`.

## Technical Skills

**Languages:** C, C++, Rust, Assembly, Java, Kotlin, JavaScript, Python, Bash, Haskell
**Frameworks:** PyTorch, Django, ReactJS, Numpy, Pandas, Tauri, LibGDX, Raylib
**Developer Tools:** Emacs, Linux, Git, GDB, Markdown, Google Cloud Platform, OpenLiteSpeed, SqlLite3
**Technical Electives:** Data Structures & Algorithms, Operating Systems, Cryptography, Database Management, Computer Security, Computer Networks, Compilers, Machine Learning, Natural Language Processing

## Awards

- Awarded the **Summer Undergraduate Research Fellowship** (2023) by **IRD-IIITD** for the project *"Utilizing Ultrasonic Distance Sensors as a Mapping Tool to Design User-Friendly CST"*.
- Awarded the **CHANAKYA Fellowship** (2024) by **iHub Anubhuti Foundation** for the project *"A Unified Approach to User Emotion Detection through Emojis and Textual Analysis"*.